



Propositional Logic

PPT PRESENTED BY

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What is Propositional Logic?

Propositional Logic is a branch of formal logic dealing with propositions and their combinations using logical connectives.

What is a Proposition?

A declarative statement that is either **TRUE** or **FALSE** — never both.

Propositions (have truth values):

" $2 + 2 = 4$ " $\rightarrow$ TRUE

"Paris is in Germany" $\rightarrow$ FALSE

NOT Propositions:

"Close the door!" (command)

"What time is it?" (question)

" $x + 1 = 5$ " (depends on x)

Notation and Key Terms

Variables:

p, q, r, s (lower-case letters for propositions)

Truth Values:

$T = \text{True} = 1$ $F = \text{False} = 0$

Compound Proposition:

Built using connectives: $\neg \wedge \vee \rightarrow \leftrightarrow$

Example:

$p =$ "It is raining"

$q =$ "It is cloudy"

$p \rightarrow q =$ "If raining, then cloudy"

Logical Connectives

Five fundamental connectives build compound propositions from atomic ones:

Symbol	Name	Read As	Example ($p = \text{'it is hot'}$, $q = \text{'it is humid'}$)
$\neg p$	Negation	NOT p	<i>"It is NOT hot"</i>
$p \wedge q$	Conjunction	p AND q	<i>"It is hot AND humid"</i>
$p \vee q$	Disjunction	p OR q (inclusive)	<i>"It is hot OR humid (or both)"</i>
$p \rightarrow q$	Conditional	If p , then q	<i>"If it is hot, then it is humid"</i>
$p \leftrightarrow q$	Biconditional	p if and only if q	<i>"Hot if and only if humid"</i>

Truth Tables

A truth table systematically evaluates a compound proposition for every possible combination of truth values:

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

★ Key Insight ($p \rightarrow q$): The conditional is FALSE only when $p = \text{TRUE}$ and $q = \text{FALSE}$ (Row 2). A FALSE hypothesis makes the implication vacuously TRUE — even 'If $1=2$ then pigs fly' is TRUE!

Logical Equivalence & Key Laws

Two propositions P and Q are logically equivalent ($P \equiv Q$) if they have identical truth values for every possible assignment of variable values.

De Morgan's Laws (most important equivalences in logic)

$\neg(p \wedge q) \equiv \neg p \vee \neg q \rightarrow$ NOT of (AND) = OR of (NOTs)

$\neg(p \vee q) \equiv \neg p \wedge \neg q \rightarrow$ NOT of (OR) = AND of (NOTs)

Other Key Equivalences:

Contrapositive: $p \rightarrow q \equiv \neg q \rightarrow \neg p$

Double Negation: $\neg(\neg p) \equiv p$

Idempotent: $p \wedge p \equiv p$ and $p \vee p \equiv p$

Implication Rewrite: $p \rightarrow q \equiv \neg p \vee q$

Applications in Computer Science

Propositional Logic is the theoretical backbone of key areas in computer science:

Digital Circuit Design

Logic gates (AND, OR, NOT, NAND, NOR, XOR) are direct hardware implementations of logical connectives. Boolean algebra — rooted in propositional logic — is used to design, analyze, and minimize digital circuits in processors, memory, and all digital devices.

Database Query Optimization

SQL WHERE clauses use AND, OR, NOT operators: WHERE (age > 18) AND (city = 'Hyderabad'). Database engines apply De Morgan's Laws and other equivalences to transform queries into faster, equivalent forms automatically.

Programming & Software Testing

Every if-else and loop condition is propositional logic in code. Software testers use truth tables to design test cases ensuring every logical combination of conditions is exercised — a technique called Condition Coverage Testing.

Artificial Intelligence

Expert systems use propositional logic for knowledge representation and rule-based inference. Automated theorem provers use Resolution Refutation — a proof technique built entirely on propositional logic — to verify program correctness.



THANK YOU

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